

The Exponent Rules (Part III)

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Four Rule Problems

Simplify each expression below and write the Roman numerals of the exponent rules that you use beside each step. Do not leave your answers with fraction exponents or negative exponents.

① $\left(\frac{x^2 \cdot x^7}{x^3}\right)^{-5}$

I $\left(\frac{x^{2+7}}{x^3}\right)^{-5}$

$\left(\frac{x^9}{x^3}\right)^{-5}$

II $(x^{9-3})^{-5}$

$(x^6)^{-5}$

V $x^{6(-5)}$

x^{-30}

VIII $\boxed{\frac{1}{x^{30}}}$

② $(a^2 \cdot y^{10} \cdot a^2)^{\frac{1}{5}}$

I $(a^{2+2} \cdot y^{10})^{\frac{1}{5}}$

$(a^4 \cdot y^{10})^{\frac{1}{5}}$

III $(a^4)^{\frac{1}{5}} (y^{10})^{\frac{1}{5}}$

V $a^{4(\frac{1}{5})} y^{10(\frac{1}{5})}$

$a^{\frac{4}{5}} y^2$

X $\boxed{(\sqrt[5]{a})^4 y^2}$

$$\textcircled{3} \left(\frac{7^3 8^0}{7^6} \right)^{-\frac{1}{6}}$$

$$\text{VII} \left(\frac{7^3(1)}{7^6} \right)^{-\frac{1}{6}}$$

$$\left(\frac{7^3}{7^6} \right)^{-\frac{1}{6}}$$

$$\text{II} (7^{3-6})^{-\frac{1}{6}}$$

$$(7^{-3})^{-\frac{1}{6}}$$

$$\text{I} 7^{-3(-\frac{1}{6})}$$

$$7^{\frac{1}{2}}$$

IX

$$\boxed{\sqrt{7}}$$

VI

I

V

VIII

$$\textcircled{4} (b \cdot b^6)^{-8}$$

$$(b^1 \cdot b^6)^{-8}$$

$$(b^{1+6})^{-8}$$

$$(b^7)^{-8}$$

$$b^{7(-8)}$$

$$b^{-56}$$

$$\boxed{\frac{1}{b^{56}}}$$

$$\textcircled{5} (3^{-7} \cdot 5^0 \cdot 3^{12})^{\frac{1}{6}}$$

$$\text{I} (3^{-7+12} \cdot 5^0)^{\frac{1}{6}}$$

$$(3^5 \cdot 5^0)^{\frac{1}{6}}$$

$$\text{VII} (3^5 \cdot 1)^{\frac{1}{6}}$$

$$(3^5)^{\frac{1}{6}}$$

$$\text{V} 3^{5(\frac{1}{6})}$$

$$3^{\frac{5}{6}}$$

$$\text{X} \boxed{(\sqrt[6]{3})^5}$$

I

II

V

VIII

$$\textcircled{6} \left(\frac{y^9}{y^2 \cdot y^3} \right)^{-2}$$

$$\left(\frac{y^9}{y^{2+3}} \right)^{-2}$$

$$\left(\frac{y^9}{y^5} \right)^{-2}$$

$$(y^{9-5})^{-2}$$

$$(y^4)^{-2}$$

$$y^{4(-2)}$$

$$y^{-8}$$

$$\boxed{\frac{1}{y^8}}$$

* You may leave your answer as a power expression

⑦ $(2^{10} \cdot 2)^{-2}$

VI $(2^{10} \cdot 2^1)^{-2}$

I $(2^{10+1})^{-2}$

$(2^{11})^{-2}$

V $2^{11(-2)}$

2^{-22}

VIII $\boxed{\frac{1}{2^{22}}}$

⑧ $(a^3 \cdot a^4 \cdot b^{34})^{\frac{2}{17}}$

I $(a^{3+4} \cdot b^{34})^{\frac{2}{17}}$

$(a^7 \cdot b^{34})^{\frac{2}{17}}$

III $(a^7)^{\frac{2}{17}} (b^{34})^{\frac{2}{17}}$

V $a^{7(\frac{2}{17})} b^{34(\frac{2}{17})}$

$a^{\frac{14}{17}} b^4$

X $\boxed{(\sqrt[17]{a})^{14} b^4}$

⑨ $\left(\frac{x^8}{y^7 \cdot y^{20}}\right)^{\frac{1}{9}}$

I $\left(\frac{x^8}{y^{7+20}}\right)^{\frac{1}{9}}$

$\left(\frac{x^8}{y^{27}}\right)^{\frac{1}{9}}$

IV $\frac{(x^8)^{\frac{1}{9}}}{(y^{27})^{\frac{1}{9}}}$

V $\frac{x^{8(\frac{1}{9})}}{y^{27(\frac{1}{9})}}$

X $\frac{x^{8/9}}{y^3} = \boxed{\frac{(\sqrt[9]{x})^8}{y^3}}$

⑩ $\left(\frac{a^2 \cdot a^{\frac{3}{5}}}{b^{30}}\right)^{\frac{1}{3}}$

I $\left(\frac{a^{2+\frac{3}{5}}}{b^{30}}\right)^{\frac{1}{3}}$

$\left(\frac{a^{\frac{13}{5}}}{b^{30}}\right)^{\frac{1}{3}}$

IV $\frac{(a^{\frac{13}{5}})^{\frac{1}{3}}}{(b^{30})^{\frac{1}{3}}}$

V $\frac{a^{\frac{13}{5}(\frac{1}{3})}}{b^{30(\frac{1}{3})}}$

$\frac{a^{\frac{13}{15}}}{b^{10}}$

X $\boxed{\frac{(\sqrt[15]{a})^{13}}{b^{10}}}$

$$\begin{array}{l} \text{VII} \quad \textcircled{11} \left(\frac{a^0 \cdot b^6}{c^3 \cdot c^6} \right)^5 \\ \left(\frac{1 \cdot b^6}{c^3 \cdot c^6} \right)^5 \\ \left(\frac{b^6}{c^3 \cdot c^6} \right)^5 \\ \text{I} \quad \left(\frac{b^6}{c^{3+6}} \right)^5 \\ \left(\frac{b^6}{c^9} \right)^5 \\ \text{IV} \quad \frac{(b^6)^5}{(c^9)^5} \\ \text{V} \quad \frac{b^{6(5)}}{c^{9(5)}} = \boxed{\frac{b^{30}}{c^{45}}} \end{array}$$

$$\begin{array}{l} \text{VI} \quad \textcircled{12} \left(\frac{z \cdot z^8}{x^4} \right)^4 \\ \left(\frac{z^1 \cdot z^8}{x^4} \right)^4 \\ \text{I} \quad \left(\frac{z^{1+8}}{x^4} \right)^4 \\ \left(\frac{z^9}{x^4} \right)^4 \\ \text{IV} \quad \frac{(z^9)^4}{(x^4)^4} \\ \text{V} \quad \frac{z^{9(4)}}{x^{4(4)}} \\ \boxed{\frac{z^{36}}{x^{16}}} \end{array}$$

More Complicated Expressions with Negative Exponents

Rule VIII') $\frac{1}{x^{-a}} = x^a$

Proof: $\frac{1}{x^{-a}} = \frac{1}{\frac{1}{x^a}} = 1 \div \frac{1}{x^a} = \frac{1}{1} \div \frac{1}{x^a} = \frac{1}{1} \cdot \frac{x^a}{1} = \frac{x^a}{1} = x^a$

Rule VIII)

A fraction symbol can be interpreted as the division operation.

Any number can be written as a fraction with "1" in its denominator.

Write the following expressions with positive exponents and simplify your final answers if possible. Write the Roman numerals of the rules that you use in the margin.

VIII' $\textcircled{13} \frac{1}{x^{-3}} = \boxed{x^3}$

VIII', X $\textcircled{14} \frac{1}{27^{-\frac{2}{3}}} = 27^{\frac{2}{3}} = (\sqrt[3]{27})^2 = (3)^2 = 3^2 = \boxed{9}$

VIII', VI $\textcircled{15} \frac{8}{b^{-1}} = 8 \cdot \frac{1}{b^{-1}} = \frac{8}{1} \cdot \frac{1}{b^{-1}} = \frac{8}{1} \cdot b^1 = \frac{8}{1} \cdot b = 8 \cdot b = \boxed{8b}$

$$\text{VIII}' \quad (16) \frac{b^7}{a^{-4}} = \boxed{b^7 a^4}$$

$$\text{VIII}' \quad (17) \frac{4x^5}{y^{-2}} = \boxed{4x^5 y^2}$$

$$\text{VIII}' \quad (18) \frac{5x^{11}}{2^{-3}} = 5x^{11} 2^3 = 5x^{11} (8) = \boxed{40x^{11}}$$

$$\text{VIII}' \quad (19) \frac{9c^6 b^4}{a^{-8} y^{-7}} = \boxed{9c^6 b^4 a^8 y^7}$$

$$\text{VIII}' \quad (20) \frac{1}{y^{-10}} = \boxed{y^{10}}$$

$$\text{VIII}', \text{IX} \quad (21) \frac{1}{100^{-\frac{1}{2}}} = 100^{\frac{1}{2}} = \sqrt{100} = \boxed{10}$$

$$\text{VIII}' \quad (22) \frac{12}{a^{-5}} = \boxed{12a^5}$$

$$\text{VIII}' \quad (23) \frac{x^2}{y^{-4}} = \boxed{x^2 y^4}$$

$$\text{VIII}' \quad (24) \frac{-3c^4}{b^{-7}} = \boxed{-3c^4 b^7}$$

$$\text{VIII}' \quad (25) \frac{-2a^3}{(-4)^{-2}} = -2a^3 (-4)^2 = -2a^3 (16) = \boxed{-32a^3}$$

$$\text{VIII}' \quad (26) \frac{8a^4 b^2}{x^{-2} y^{-5} 3^{-2}} = 8a^4 b^2 x^2 y^5 3^2 = 8a^4 b^2 x^2 y^5 (9) = \boxed{72a^4 b^2 x^2 y^5}$$

$$\text{VIII}' \quad (27) \frac{y^7}{6x^{-8}} = \boxed{\frac{y^7 x^8}{6}}$$

$$\text{VIII}' \quad (28) \frac{10b^6}{9x^{-3} y^2} = \boxed{\frac{10b^6 x^3}{9y^2}}$$

$$\text{VIII}' \quad (29) \frac{b^4}{5^{-3} c^6} = \frac{b^4 5^3}{c^6} = \frac{b^4 (125)}{c^6} = \boxed{\frac{125b^4}{c^6}}$$

$$\text{VIII}' \quad (30) \quad \frac{2x^9}{7a^3b^{-6}} = \boxed{\frac{2x^9b^6}{7a^3}}$$

$$\begin{aligned} \text{VIII}, \text{VIII}' \quad (31) \quad \frac{a^{-2}b^6}{x^{-5}y^7} &= \frac{a^{-2}}{x^{-5}} \cdot \frac{b^6}{y^7} = a^{-2} \cdot \frac{1}{x^{-5}} \cdot \frac{b^6}{y^7} \\ &= \frac{1}{a^2} \cdot x^5 \cdot \frac{b^6}{y^7} \\ &= \frac{1}{a^2} \cdot \frac{x^5}{1} \cdot \frac{b^6}{y^7} \\ &= \boxed{\frac{x^5b^6}{a^2y^7}} \end{aligned}$$

$$\text{VIII}', \text{VIII} \quad (32) \quad \frac{8x^7y^{-3}}{4a^{-7}c^2} = \boxed{\frac{2x^7a^7}{c^2y^3}}$$

$$\text{VIII}', \text{VIII} \quad (33) \quad \frac{3c^{-10}}{9b^{-6}} = \boxed{\frac{b^6}{3c^{10}}}$$

$$\text{VIII}', \text{VIII} \quad (34) \quad \frac{10x^{-2}y^{-9}}{15a^{-4}b^{-8}} = \boxed{\frac{2a^4b^8}{3x^2y^9}}$$

$$\text{VIII}', \text{VIII} \quad (35) \quad \frac{x^5y^{-3}}{2c^4b^{-8}} = \boxed{\frac{x^5b^8}{2c^4y^3}}$$

$$\text{VIII}, \text{VIII}' \quad (36) \quad \frac{56a^{-4}b^3}{7x^{-9}y^5} = \boxed{\frac{8x^9b^3}{a^4y^5}}$$

$$\text{VIII}', \text{VIII} \quad (37) \quad \frac{4a^{-11}b}{12x^{-2}c} = \boxed{\frac{x^2b}{3a^{11}c}}$$

$$\begin{aligned} \text{VIII}, \text{VIII}' \\ \text{VI} \quad (38) \quad \frac{8b^{-8}c^{-2}}{6a^{-2}x^{-1}} &= \frac{4a^2x^1}{3b^8c^2} = \boxed{\frac{4a^2x}{3b^8c^2}} \end{aligned}$$

$$\text{VIII}', \text{VI} \quad (39) \quad \frac{49x^3y^8}{7c^{-1}b^{-3}} = 7x^3y^8c^1b^3 = \boxed{7x^3y^8cb^3}$$

$$\text{VIII} \quad (40) \quad \frac{6x^{-4}y^{-3}}{48a^3b^6} = \boxed{\frac{1}{8a^3b^6x^4y^3}}$$

$$\text{VIII}' \quad (41) \frac{6a^7}{3b^{-4}} = \boxed{2a^7b^4}$$

$$\text{VIII, VI} \quad (42) \frac{4x^{-1}b^{-9}}{16y^3} = \frac{1}{4y^3x^1b^9} = \boxed{\frac{1}{4y^3xb^9}}$$

$$\text{VIII, IV} \quad (43) \left(\frac{y^{-2}}{4}\right)^2 = \left(\frac{1}{4y^2}\right)^2 = \frac{(1)^2}{(4y^2)^2} = \frac{1}{4^2(y^2)^2} = \frac{1}{16y^{2(2)}} \\ \text{III, V} \quad = \boxed{\frac{1}{16y^4}}$$

$$\text{VIII}', \text{III} \quad (44) \left(\frac{x^6}{3^{-2}}\right)^2 = (3^2x^6)^2 = (9x^6)^2 = 9^2(x^6)^2 \\ \text{V} \quad = 81x^{6(2)} \\ = \boxed{81x^{12}}$$

$$\text{VIII, IV} \quad (45) \left(\frac{2^{-2}}{x^4}\right)^3 = \left(\frac{1}{2^2x^4}\right)^3 = \left(\frac{1}{4x^4}\right)^3 = \frac{(1)^3}{(4x^4)^3} = \frac{1}{4^3(x^4)^3} = \frac{1}{64x^{4(3)}} \\ \text{III, V} \quad = \boxed{\frac{1}{64x^{12}}}$$

$$\text{VIII}', \text{VI} \quad (46) \left(\frac{a^3}{4^{-1}}\right)^3 = (4^1a^3)^3 = (4a^3)^3 = 4^3(a^3)^3 = 64a^{3(3)} \\ \text{III, V} \quad = \boxed{64a^9}$$

$$\text{VIII} \quad (47) \left(\frac{3}{5}\right)^{-2}$$

$$\text{Method I: } \frac{1}{\left(\frac{3}{5}\right)^2} = \frac{1}{\frac{3}{5} \cdot \frac{3}{5}} = \frac{1}{\frac{9}{25}} = 1 \div \frac{9}{25} = \frac{1}{1} \div \frac{9}{25} \\ = \frac{1}{1} \cdot \frac{25}{9} = \frac{25}{9}$$

$$\text{Method II: } \frac{3^{-2}}{5^{-2}} = \frac{5^2}{3^2} = \boxed{\frac{25}{9}} \\ \text{IV, VIII} \quad \text{VIII}'$$

$$\text{VIII, VI} \quad (48) \left(\frac{1}{2}\right)^{-1} = \frac{1}{\left(\frac{1}{2}\right)^1} = \frac{1}{\frac{1}{2}} = 1 \div \frac{1}{2} = \frac{1}{1} \div \frac{1}{2} = \frac{1}{1} \cdot \frac{2}{1} \\ = \frac{2}{1} \\ = \boxed{2}$$

$$\text{VIII} \quad (49) \left(\frac{4}{5}\right)^{-2} = \frac{1}{\left(\frac{4}{5}\right)^2} = \frac{1}{\frac{4}{5} \cdot \frac{4}{5}} = \frac{1}{\frac{16}{25}} = 1 \div \frac{16}{25} = \frac{1}{1} \div \frac{16}{25} \\ = \frac{1}{1} \cdot \frac{25}{16} = \boxed{\frac{25}{16}}$$

VIII, IV ⑤⑩ $\left(\frac{2}{3}\right)^{-4} = \frac{1}{\left(\frac{2}{3}\right)^4} = \frac{1}{\frac{2^4}{3^4}} = \frac{1}{\frac{16}{81}} = 1 \div \frac{16}{81} = \frac{1}{1} \div \frac{16}{81}$
 $= \frac{1}{1} \cdot \frac{81}{16}$
 $= \boxed{\frac{81}{16}}$

Four Rule Problems Requiring Rule X'

⑤⑪ $(\sqrt{9^2 \cdot 9^2})^{-1}$
 I $(\sqrt{9^{2+2}})^{-1}$
 $(\sqrt{9^4})^{-1}$
 X' $[(\sqrt{9})^4]^{-1}$
 $[(3)^4]^{-1}$
 VIII, VI $[81]^{-1} = 81^{-1} = \frac{1}{81^1} = \boxed{\frac{1}{81}}$

⑤⑫ $\left(\sqrt[3]{\frac{27^6}{27^2}}\right)^{-\frac{1}{2}}$
 II $(\sqrt[3]{27^{6-2}})^{-\frac{1}{2}}$
 $(\sqrt[3]{27^4})^{-\frac{1}{2}}$
 X' $[(\sqrt[3]{27})^4]^{-\frac{1}{2}}$
 $[3^4]^{-\frac{1}{2}}$
 $[81]^{-\frac{1}{2}}$
 $81^{-\frac{1}{2}}$
 VIII $\frac{1}{81^{\frac{1}{2}}}$
 IX $\frac{1}{\sqrt{81}} = \boxed{\frac{1}{9}}$

⑤⑬ $(\sqrt[4]{16 \cdot 16^2})^{-2}$
 VI $(\sqrt[4]{16^1 \cdot 16^2})^{-2}$
 I $(\sqrt[4]{16^{1+2}})^{-2}$
 $(\sqrt[4]{16^3})^{-2}$
 X' $[(\sqrt[4]{16})^3]^{-2}$
 $[2^3]^{-2}$
 $[8]^{-2}$
 8^{-2}
 VIII $\frac{1}{8^2}$
 $\boxed{\frac{1}{64}}$